

Nsak as a Framework for Agentic-AI enhanced Network Security

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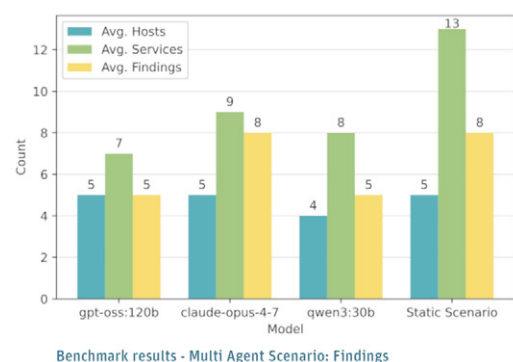
This thesis investigates whether Agentic-AI can effectively automate and enhance network security scenarios within the NSAK framework, a modular Python-based cybersecurity orchestration platform for purple teams.

Introduction

Red and blue team exercises are established cybersecurity practices, yet they remain largely inaccessible to small- and medium-sized organizations due to the expertise and recurring effort these practices demand. We present an approach that combines **LLMs** with existing network security tools to reduce knowledge barriers and time investment. Concretely, we integrated an **AI agent** with tool-calling capabilities into the **NSAK framework** - a network security test automation framework established in our preceding work - and developed an agentic network reconnaissance scenario. We evaluated its effectiveness across models of varying size, benchmarking it against a scripted scenario in a reproducible network environment.

Technical Specification

The agent is implemented using **LangChain**, an open-source agent framework. Three models are benchmarked across n runs in a virtual container lab and the BFH network security lab: a frontier model, claude-opus-4-7; an institutionally hosted open-weight model, gpt-oss:120b; and a locally hosted open-source model, qwen3:30b, via Ollama. The evaluation further compares **single-** and **multi-agent** configurations, including the use of **structured output** to ensure consistent and parsable responses.

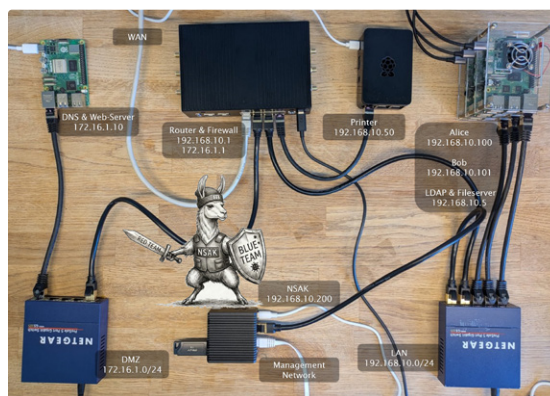


Conclusion

The results show that even smaller models can autonomously perform standard red team activities. The **AI-driven approach** demonstrates a clear advantage over scripted counterparts. The agent assesses results, generates reports with prioritized findings, and classifies steps to address vulnerabilities. This work highlights the emerging capabilities of LLMs, making cybersecurity use cases more accessible and cost-effective.

Index Terms:

AI-Driven Cybersecurity, Large Language Models, Network Security Automation Framework



Physical-Environment



Frank Erich Gauss



Lukas Christian von Allmen